ASSIGNMENT 1: LINUX BASICS

1. Creating and Renaming Files/Directories

1. Create a directory named test_dir using `mkdir`.
2. Inside test_dir, create an empty file called `example.txt`.
3. Rename example.txt to renamed_example.txt using `mv`.

```
[rohitbhupat@Rohits-MacBook-Air Documents % cd sahil_devops
[rohitbhupat@Rohits-MacBook-Air sahil_devops %
[rohitbhupat@Rohits-MacBook-Air sahil_devops % mkdir test_dir
[rohitbhupat@Rohits-MacBook-Air sahil_devops %
[rohitbhupat@Rohits-MacBook-Air sahil_devops % cd test_dir
[rohitbhupat@Rohits-MacBook-Air test_dir %
[rohitbhupat@Rohits-MacBook-Air test_dir % touch example.txt
[rohitbhupat@Rohits-MacBook-Air test_dir %
[rohitbhupat@Rohits-MacBook-Air test_dir % ls -ltr
total 0
-rw-r--r--  1 rohitbhupat  staff  0 16 Oct 10:41 example.txt
[rohitbhupat@Rohits-MacBook-Air test_dir %
[rohitbhupat@Rohits-MacBook-Air test_dir %
[rohitbhupat@Rohits-MacBook-Air test_dir % mv example.txt renamed_example.txt
[rohitbhupat@Rohits-MacBook-Air test_dir %
[rohitbhupat@Rohits-MacBook-Air test_dir % ls -ltr
total 0
-rw-r--r--  1 rohitbhupat  staff  0 16 Oct 10:41 renamed_example.txt
```

Explanation:

- `mkdir test_dir`: Creates a new directory named `test_dir`.
- `cd test_dir`: Navigates into the newly created directory.
- `touch example.txt`: Creates an empty file named `example.txt`.
- `ls -ltr`: Displays the files in the current directory with details such as permissions, owner, size, and modification time.
- `mv example.txt renamed_example.txt`: Renames the file from `example.txt` to `renamed_example.txt`.

Result: The directory was created successfully, an empty file was created inside it, and the file was renamed to `renamed_example.txt`. The terminal output confirms that the renamed file exists.

2. Viewing File Contents

1. Use `cat` to display the contents of /etc/passwd.
2. Display only the first 5 lines of /etc/passwd using `head`.
3. Display only the last 5 lines of /etc/passwd using `tail`.

```
rohitbhupat@Rohits-MacBook-Air test_dir % cat /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode.  At other times this information is provided by
# Open Directory.
#
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
nobody:*:-2:-2:Unprivileged User:/var/empty:/usr/bin/false
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_uucp:*:4:4:Unix to Unix Copy Protocol:/var/spool/uucp:/usr/sbin/uucico
_taskgated:*:13:13:Task Gate Daemon:/var/empty:/usr/bin/false
_networkd:*:24:24:Network Services:/var/networkd:/usr/bin/false
_installassistant:*:25:25:Install Assistant:/var/empty:/usr/bin/false
_lp:*:26:26:Printing Services:/var/spool/cups:/usr/bin/false
_postfix:*:27:27:Postfix Mail Server:/var/spool/postfix:/usr/bin/false
_scsd:*:31:31:Service Configuration Service:/var/empty:/usr/bin/false
_ces:*:32:32:Certificate Enrollment Service:/var/empty:/usr/bin/false
_appstore:*:33:33:Mac App Store Service:/var/db/appstore:/usr/bin/false
_mcxalr:*:54:54:MCX AppLaunch:/var/empty:/usr/bin/false
_appleevents:*:55:55:AppleEvents Daemon:/var/empty:/usr/bin/false
```

```
rohitbhupat@Rohits-MacBook-Air test_dir % tail -5 /etc/passwd
_spinandd:*:305:305:SPINAND Daemon:/var/db/spinandd:/usr/bin/false
_corespeechd:*:306:306:CoreSpeech Services:/var/empty:/usr/bin/false
_diagnosticservicesd:*:307:307:Diagnostic Services:/var/empty:/usr/bin/false
_mds_stores:*:308:308:Spotlight File Metadata Index Daemon:/var/empty:/usr/bin/false
_oahd:*:441:441:OAH Daemon:/var/empty:/usr/bin/false
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir % head -5 /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode.  At other times this information is provided by
rohitbhupat@Rohits-MacBook-Air test_dir %
```

Explanation:

- `cat /etc/passwd`: Displays the contents of the `/etc/passwd` file, which contains user account information on the system.
- `head -n 5 /etc/passwd`: Displays the first five lines of the file.
- `tail -n 5 /etc/passwd`: Displays the last five lines of the file.

Result: The commands display the complete file contents, the first five lines, and the last five lines, respectively. This demonstrates how Linux commands can be used to inspect text files.

3. Searching for Patterns

1. Use `grep` to find all lines containing the word “root” in /etc/passwd.

```
rohitbhupat@Rohits-MacBook-Air test_dir % grep "root" /etc/passwd
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_cvmsroot:*:212:212:CVMS Root:/var/empty:/usr/bin/false
rohitbhupat@Rohits-MacBook-Air test_dir %
```

Explanation: The `grep` command searches for lines containing the word `root` in `/etc/passwd`.

Result: The terminal displays the lines that match the search pattern.

4. Zipping and Unzipping

1. Compress the test_dir directory into a file named test_dir.zip using `zip`.
2. Unzip test_dir.zip into a new directory named `unzipped_dir`.

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % zip -r test_dir.zip test_dir
updating: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr
total 8
drwxr-xr-x  2 rohitbhupat  staff   64 16 Oct 10:33 Linux_Basics_Sahil_Shaikh
drwxr-xr-x  3 rohitbhupat  staff   96 16 Oct 10:41 test_dir
-rw-r--r--  1 rohitbhupat  staff  352 16 Oct 10:47 test_dir.zip
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

Explanation: The `zip -r` command compresses the directory and its contents into `test_dir.zip`. The `unzip` command extracts the archive into the specified `unzipped_dir` directory.

Result: The directory contents are stored in a ZIP archive and extracted into a new location.

5. Downloading Files

1. Use `wget` to download a file from a URL (e.g., https://example.com/sample.txt).

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % wget https://www.gnu.org/licenses/gpl-3.0.txt

--2026-10-17 13:21:04--  https://www.gnu.org/licenses/gpl-3.0.txt
Resolving www.gnu.org (www.gnu.org)... 209.51.188.116
Connecting to www.gnu.org (www.gnu.org)|209.51.188.116|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 35149 (34K) [text/plain]
Saving to: 'gpl-3.0.txt'

gpl-3.0.txt                      100%[=============================================================>]  34.33K   157KB/s    in 0.2s

2026-10-17 13:21:06 (157 KB/s) - 'gpl-3.0.txt' saved [35149/35149]

rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr
total 80
-rw-r--r--@ 1 rohitbhupat  staff  35149 30 Sep  2017 gpl-3.0.txt
drwxr-xr-x  2 rohitbhupat  staff     64 16 Oct 10:33 Linux_Basics_Sahil_Shaikh
-rw-r--r--  1 rohitbhupat  staff    352 16 Oct 10:47 test_dir.zip
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:48 test_dir
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:50 unzipped_dir
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

Explanation: The `wget` command downloads a file from a specified URL and saves it locally, provided the URL points to an accessible file.

Result: The download completes if the URL is valid and the server provides the file.

6. Changing Permissions

1. Create a file named secure.txt and change its permissions to read-only for everyone using `chmod`.

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % touch secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr
total 80
-rw-r--r--@ 1 rohitbhupat  staff  35149 30 Sep  2017 gpl-3.0.txt
drwxr-xr-x  2 rohitbhupat  staff     64 16 Oct 10:33 Linux_Basics_Sahil_Shaikh
-rw-r--r--  1 rohitbhupat  staff    352 16 Oct 10:47 test_dir.zip
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:48 test_dir
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:50 unzipped_dir
-rw-r--r--@ 1 rohitbhupat  staff      0 17 Oct 13:22 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr | grep "secure.txt"
-rw-r--r--@ 1 rohitbhupat  staff      0 17 Oct 13:22 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % chmod 444 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr | grep secure.txt
-r--r--r--@ 1 rohitbhupat  staff      0 17 Oct 13:22 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

Explanation: The `chmod 444` command gives the file read permission to the owner, group, and others, while removing write and execute permissions.

Result: The file becomes read-only according to its permission bits. Note that the file's owner or an administrator may still be able to change its permissions.

7. Working with Environment Variables

1. Use `export` to set a new environment variable called MY_VAR with the value “Hello, Linux!”.

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % echo MY_VAR
MY_VAR
rohitbhupat@Rohits-MacBook-Air sahil_devops % echo "$MY_VAR"
Hello
```

Explanation: The `export` command creates or updates the `MY_VAR` environment variable and makes it available to child processes launched from that shell. The `echo` command displays its value.

Result: The terminal displays `Hello!`, confirming that the variable has been set in the current shell.